

ARTIFICIAL INTELLIGENCE

A Semantic Score Generation Method for Health Misinformation Detection

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Abstract

Online health misinformation poses a significant and growing threat to public health. Existing fact-checking systems that rely on Large Language Models (LLMs) suffer from hallucination — the generation of plausible but factually incorrect content — while conventional Retrieval-Augmented Generation (RAG) systems are limited by their reliance on a single knowledge base containing only verified true claims, leaving them unable to model the structural characteristics of false information.

This paper presents a novel Dual Knowledge-Base Graph-Based Retrieval-Augmented Semantic Score Generation framework for automated health misinformation detection. Our system simultaneously compares an incoming claim against two separate knowledge bases: a Verified True Claims base and a Verified False Claims base, both encoded using Sentence-BERT. The difference between the two cosine similarity scores yields a signed distance metric — KB_Diff — that quantifies the claim's proximity to truth versus falsity. This signed distance, combined with Natural Language Inference (NLI) logical relationship scores, Semantic Health Knowledge Graph (SHKG) PageRank authority features, and linguistic pattern signals, forms a 12-dimensional hybrid feature vector. A Bagged AdaBoost classifier trained on this vector produces the final True/False decision.

To address the natural class imbalance in health claim datasets (2:1 True-to-False ratio), we employ SVM-SMOTE for boundary-aware synthetic minority oversampling followed by Tomek Links cleaning. An ablation study isolating the effect of class imbalance reveals that balanced sampling increases False-class recall from 0.067 to 0.350 — a 5.2x improvement — while improving AUC from 0.715 to 0.731. On the PUBHEALTH benchmark, our method achieves Macro-F1 of 0.580 and AUC of 0.731 under balanced evaluation, compared to 0.333 and 0.491 for the reference TrumorGPT single-KB baseline. The entire system runs on open-source components with zero inference cost, making it lightweight, reproducible, and auditable.

1. Introduction

The proliferation of health misinformation on social media and news platforms has become a critical public health challenge. False claims about vaccines, treatments, and disease transmission spread at scale — reaching millions of users within minutes — while expert fact-checking remains a slow, manual, and resource-intensive process. The COVID-19 pandemic demonstrated the lethal consequences of health misinformation, making automated detection systems an urgent research priority.

Large Language Models (LLMs) such as GPT-4, LLaMA, and Claude have demonstrated remarkable language understanding capabilities, yet they are prone to hallucination: generating contextually fluent but factually incorrect content with high apparent confidence. In medical domains, such hallucinations pose direct risks to patient safety. Furthermore, closed-source LLM APIs introduce reproducibility concerns, as model outputs may change across updates, and cost barriers that limit deployment in resource-constrained settings.

Retrieval-Augmented Generation (RAG) systems partially address the hallucination problem by grounding predictions in externally retrieved evidence. However, existing RAG approaches — including TrumorGPT, the state-of-the-art GraphRAG system for health fact-checking — rely exclusively on a single knowledge base of verified true claims. This creates two compounding problems that we term the Accuracy Paradox and Majority Blindness.

The Accuracy Paradox arises from natural class imbalance in health datasets: in the PUBHEALTH dataset, true claims outnumber false claims at a 2:1 ratio. A naive classifier that labels every claim as 'True' achieves approximately 66% accuracy without detecting any misinformation. This high accuracy figure masks complete failure at the task that matters — catching false claims. Majority Blindness is the consequence: classifiers optimized for overall loss sacrifice the minority False class entirely, yielding a False-class recall that collapses to near zero (0.067 in our imbalanced baseline). In health misinformation contexts, this is not merely a statistical problem — a missed false claim can translate directly into patient harm.

The structural challenge runs deeper than class imbalance alone. 3D Principal Component Analysis (PCA) of the 12-dimensional feature space reveals that false claims do not form a distinct, separable cluster in semantic space. Instead, they are deeply embedded within the cloud of true claims — because misinformation deliberately mimics the lexical and structural patterns of legitimate medical content, using the same entities (FDA, COVID, vaccine) in deceptive contexts. Linear Discriminant Analysis (LDA) separability on the training feature space is approximately 0.68, confirming significant class overlap that renders simple threshold-based classifiers inadequate.

Retrieval-Augmented Generation (RAG) systems partially address the hallucination problem by grounding predictions in externally retrieved evidence. However, existing single-KB RAG approaches can only measure how similar a claim is to the truth — they cannot measure how similar it is to known falsehoods. As demonstrated in our experiments, this architectural blind spot is severe: the single-KB TrumorGPT baseline achieves a False-class recall of 0.000 on balanced data, meaning it fails to detect any false health claims.

This paper makes the following contributions:

- Dual-KB Retrieval with Signed Distance: We introduce a dual knowledge-base architecture that simultaneously retrieves from both True and False knowledge bases. The KB_Diff signed distance metric encodes directional proximity to truth versus falsity as an explicit, learnable feature.
- Semantic Health Knowledge Graph (SHKG) with PageRank Authority: We construct a medical entity co-occurrence graph from training data and use PageRank centrality to derive structural reliability signals that complement text-level similarity features.
- 12-Dimensional Hybrid Feature Vector: We integrate semantic (Sentence-BERT + NLI), graph (PageRank), and linguistic (hedge words, sensational terms, numerics) signals into a unified, interpretable feature representation.
- Class Imbalance Ablation Study: We systematically study the effect of natural class imbalance using SVM-SMOTE and Tomek Links resampling, holding pipeline and dataset domain constant to isolate the imbalance variable.
- Open-Source, Reproducible, Auditable System: All components use pinned open-source libraries with zero inference cost and full per-prediction explainability via feature importance analysis.

2. Related Work

2.1 Large Language Models for Fact-Checking

Large Language Models have been applied to automated fact-checking with considerable success. GPT-based systems have demonstrated strong performance on political claim verification datasets such as PolitiFact and LIAR. However, as documented in multiple studies, LLMs are susceptible to hallucination, particularly in specialized domains such as medicine, where their training data may be incomplete or outdated. The TrumorGPT paper (Hang et al., 2025) reports strong performance using GPT-4 on PolitiFact, but acknowledges the cost and reproducibility limitations of closed-source APIs. Our work eliminates GPT dependency while demonstrating competitive performance on the open PUBHEALTH benchmark.

2.2 Retrieval-Augmented Generation

RAG architectures, introduced by Lewis et al. (2020), ground language model generation in externally retrieved documents. Standard dense retrieval uses bi-encoder models such as DPR or Sentence-BERT to encode queries and documents into a shared vector space, enabling efficient nearest-neighbor retrieval. Graph-based extensions — GraphRAG — additionally model entity relationships through knowledge graphs. TrumorGPT (Hang et al., 2025) employs Topic-Specific TextRank (TST) for sentence centrality scoring and Jaccard similarity for triple matching. Our work extends this framework by introducing a second, complementary false-claims knowledge base and replacing Jaccard matching with dense Sentence-BERT cosine similarity, yielding substantially higher discriminative power.

2.3 Graph-Based Representations for Health NLP

Knowledge graphs have been widely used in biomedical NLP to represent entity relationships extracted from clinical text. PageRank-based centrality measures have been employed in TextRank (Mihalcea & Tarau, 2004) for extractive summarization and keyword extraction. We adapt this paradigm to construct a Semantic Health Knowledge Graph (SHKG) from claim-level entity co-occurrences, where PageRank provides a data-driven authority score for medical concepts. Unlike pre-built biomedical ontologies (UMLS, MeSH), our SHKG is constructed dynamically from the training data, ensuring domain alignment without external dependencies.

2.4 Class Imbalance in Misinformation Detection

Class imbalance is a pervasive challenge in misinformation detection, where genuine false claims are typically rarer than true or unverified claims. Standard SMOTE (Chawla et al., 2002) addresses imbalance through random interpolation in the minority class, but can generate noisy samples in overlapping regions. SVM-SMOTE (Nguyen et al., 2009) improves upon this by restricting synthetic sample generation to the decision boundary region identified by a Support Vector Machine. Tomek Links (Tomek, 1976) removes majority-class samples that are nearest neighbors to minority-class samples, sharpening the decision boundary. To our knowledge, no prior health misinformation detection system has systematically studied the effect of class imbalance through a controlled ablation study holding pipeline and domain constant.

3. Proposed Approach

3.1 System Overview

Our pipeline consists of six sequential stages: (1) health claim input, (2) named entity extraction via spaCy, (3) dual knowledge-base retrieval using Sentence-BERT cosine similarity, (4) SHKG-based PageRank graph scoring, (5) 12-dimensional hybrid feature vector construction, and (6) Bagged AdaBoost classification. Unlike classical RAG systems that produce text in the generation step, our system produces interpretable numerical semantic scores — hence the designation Semantic Score Generation.

3.2 Feature Space Analysis: Overlap and Entanglement

Before describing the retrieval architecture, it is instructive to understand why standard text-matching approaches fail. 3D PCA visualization of the 12-dimensional feature space reveals that false claims do not form a geometrically separable cluster. Instead, false claims (red) are deeply embedded within the cloud of true claims (green), with substantial overlap in all three principal components. This structural entanglement — which we term feature space overlap — arises because misinformation deliberately borrows the vocabulary, entities, and sentence structure of legitimate medical content.

LDA separability analysis on the training feature space yields a score of approximately 0.68, indicating that the true and false classes are not linearly separable in the raw feature representation. This result motivates three design decisions: the use of a non-linear ensemble classifier (Bagged AdaBoost over decision trees), boundary-aware resampling (SVM-SMOTE), and the dual-KB architecture that generates a signed distance signal explicitly encoding which class a claim is structurally closer to.

3.3 Dual Knowledge-Base Retrieval

Given an input claim c, we maintain two knowledge bases constructed from training data:

- KB_True: the set of verified true health claims with their Sentence-BERT embeddings.
- KB_False: the set of verified false health claims with their Sentence-BERT embeddings.

Both knowledge bases are encoded using the all-MiniLM-L6-v2 Sentence-BERT model and L2-normalized. For a query claim c, the cosine similarity to each knowledge base is computed as:

$$\mathbf{Cos_{true} = max_i \{ q \cdot kb_t_i \} \quad (L2\text{-}normalized\ vectors)}$$

$$\mathbf{Cos_{false} = max_j \{ q \cdot kb_f_j \} \quad (L2\text{-}normalized\ vectors)}$$

The signed distance metric KB_Diff is then defined as:

$$\mathbf{KB_Diff = Cos_{true} - Cos_{false}}$$

A positive KB_Diff indicates the claim is semantically closer to verified truth; a negative value indicates proximity to known falsehoods. This signed distance encodes both direction and magnitude of the claim's position in the semantic space, constituting an explicit, learnable feature that no single-KB system can produce. To prevent self-match leakage during training,

each training claim's own position in the knowledge base is masked with a similarity score of -1.0 before retrieval.

3.4 Natural Language Inference Layer

Cosine similarity measures semantic similarity but cannot detect logical contradiction. For example, 'Vaccines are safe' and 'Vaccines are harmful' share high cosine similarity due to lexical overlap, yet are logically contradictory. We address this by passing each retrieved context-claim pair through an NLI model (cross-encoder/nli-deberta-v3-base). NLI is applied twice per claim — once with the best-matching True KB context, once with the best-matching False KB context — yielding four features: sup_true, con_true, sup_false, con_false.

3.5 Semantic Health Knowledge Graph and PageRank

The SHKG is constructed as a weighted co-occurrence graph from training claims exclusively, preventing test data leakage. Nodes represent medical entities extracted by spaCy (types: ORG, GPE, PERSON, PRODUCT). An edge between entities u and v is added for each training claim in which both entities appear, with edge weight incremented by 1 for each co-occurrence. PageRank centrality is computed with damping factor d = 0.85. For a given claim, the graph_score is:

$$\text{graph_score}(c) = \text{mean}\ \{\ PR(e)\ :\ e \in \text{entities}(c)\ \}$$

This score captures structural authority — highly interconnected medical concepts such as FDA, WHO, and COVID receive high PageRank scores, reflecting their centrality in the medical literature. This structural signal is complementary to and independent of text-level cosine similarity.

3.6 Linguistic Feature Extraction

Four linguistic signals capture the writing style of misinformation:

- hedge_score: density of epistemic uncertainty markers (e.g., 'might', 'allegedly', 'reportedly').
- sens_score: density of sensationalist lexemes (e.g., 'miracle', 'secret', 'shocking', 'cure').
- num_score: density of numerical expressions.
- question_flag: binary indicator of question marks.

Additionally, wc_norm is the word count normalized by the training maximum. Feature importance analysis (Section 5.3) confirms that wc_norm is the single most discriminative feature: false health claims exhibit a characteristic bimodal length distribution — either very short clickbait or excessively verbose pseudo-scientific text mimicking legitimate research.

3.7 Classification

The 12-dimensional feature vector undergoes StandardScaler normalization before classification. We employ a two-level ensemble: AdaBoostClassifier (n_estimators=50) operating over DecisionTreeClassifier base learners, wrapped in a BaggingClassifier (n_estimators=10, max_samples=0.8). Bagging reduces variance and prevents overfitting on

the small dataset; AdaBoost concentrates learning effort on the decision boundary region where false and true claims are most similar — precisely the entangled region identified by PCA analysis. The classification threshold is optimized for macro-F1 using a held-out validation slice of the training data, calibrated via the Precision-Recall curve.

3.8 Class Imbalance Handling

The PUBHEALTH dataset exhibits a natural 2:1 True-to-False ratio. We apply SVM-SMOTE (k_neighbors=5, m_neighbors=10, svm_estimator=SVC(kernel='rbf')) to generate synthetic False samples restricted to the SVM-identified decision boundary region. SVM-SMOTE's boundary-awareness prevents the generation of noisy samples in regions dominated by the majority class — a known failure mode of standard SMOTE that would worsen the already entangled feature space. Tomek Links cleaning (sampling_strategy='majority') then removes majority-class samples that are nearest neighbors to minority-class samples, sharpening the decision boundary. Both operations are applied exclusively to training data.

4. Experimental Setup

4.1 Dataset

We use the PUBHEALTH dataset (Kotonya & Toni, 2020), a publicly available health fact-checking benchmark. Raw labels include: true, mostly true, mixture, false, pants-on-fire, and unproven. We map true and mostly true to label 1 (True) and false and pants-on-fire to label 0 (False). Ambiguous categories (mixture, unproven) are excluded as they do not provide clean binary supervision. Two experimental configurations are evaluated:

- Imbalanced (2:1): 600 True and 300 False claims sampled with random_state=42. Train/test split: 720/180 (stratified 80/20).
- Balanced (1:1): 300 True and 300 False claims sampled with random_state=99. Train/test split: 480/120 (stratified 80/20).

Both configurations use the same source dataset and pipeline, differing only in class sampling ratio. This design isolates the effect of class imbalance at evaluation time while holding domain and pipeline constant.

4.2 Baselines

We compare against the TrumorGPT reference system (Hang et al., 2025) re-implemented on PUBHEALTH for fair comparison. The baseline uses Topic-Specific TextRank (TST) for sentence centrality, Jaccard similarity for triple matching, and a threshold-based classifier. We evaluate the baseline under both balanced and imbalanced configurations, with and without SVM-SMOTE resampling, producing four total comparison configurations across the two methods.

4.3 Evaluation Metrics

Given the class imbalance problem, we report four metrics:

- Accuracy: overall correct predictions. Reported for context but acknowledged as misleading on imbalanced data due to the Accuracy Paradox.
- Macro-F1: harmonic mean of per-class F1 scores with equal class weighting, the primary metric for imbalanced evaluation.
- Recall (False): True Positive Rate for the minority False class — the most clinically critical metric.
- AUC-ROC: area under the receiver operating characteristic curve, measuring discrimination power independent of threshold choice.

4.4 Implementation Details

All experiments run on a free Google Colab session (CPU for classification, GPU available for NLI inference). Key components: Sentence-BERT all-MiniLM-L6-v2 for dense encoding, cross-encoder/nli-deberta-v3-base for NLI, spaCy en_core_web_sm for entity extraction, NetworkX for SHKG construction and PageRank computation, scikit-learn for classification, and imbalanced-learn for SVM-SMOTE and Tomek Links. Downstream classifier inference latency is approximately 2ms per claim on a single CPU thread; note that full pipeline latency including SBERT encoding and NLI inference is 0.5-2 seconds per claim and is dominated by the NLI model.

5. Performance Analysis

5.1 Main Results

Table 1 presents the full comparison across four configurations. Two key comparisons are highlighted:

Comparison 1 — Balanced dataset, no resampling (Row 1 vs. Row 3): The TrumorGPT baseline achieves Recall(False) = 0.000, meaning it detects no false claims on balanced data. AUC of 0.491 is indistinguishable from random guessing. Our Dual-KB method achieves Recall(False) = 0.350 and AUC = 0.731, with Macro-F1 of 0.580 versus 0.333 for the baseline — a 74% relative improvement in Macro-F1.

Comparison 2 — Imbalanced dataset with SVM-SMOTE (Row 2 vs. Row 4): The baseline achieves Recall(False) = 0.983 by over-indexing on the minority class at the expense of overall accuracy (0.328). Our method maintains Accuracy of 0.661 with Macro-F1 of 0.453, achieving a more calibrated balance between precision and recall across both classes.

Method	Dataset	Resampling	Accuracy	F1 (macro)	Recall (False)	AUC-ROC
TrumorGPT (TST+Centrality+Jaccard)	Balanced (1:1)	No	0.500	0.333	0.000	0.491
TrumorGPT (TST+Centrality+Jaccard+SMOTE+Bagging)	Imbalanced (2:1)	SVM-SMOTE + TomekLinks	0.328	0.247	0.983	0.496

Method	Dataset	Resampling	Accuracy	F1 (macro)	Recall (False)	AUC-ROC
Novelty Method (Dual-KB)	Balanced (1:1)	No	0.608	0.580	0.350	0.731
Novelty Method (Dual-KB+SMOTE+Bagging)	Imbalanced (2:1)	SVM-SMOTE + TomekLinks	0.661	0.453	0.067	0.715
TrumorGPT GPT-4 (paper original)*	PolitiFact 600 claims	—	0.885	—	—	—

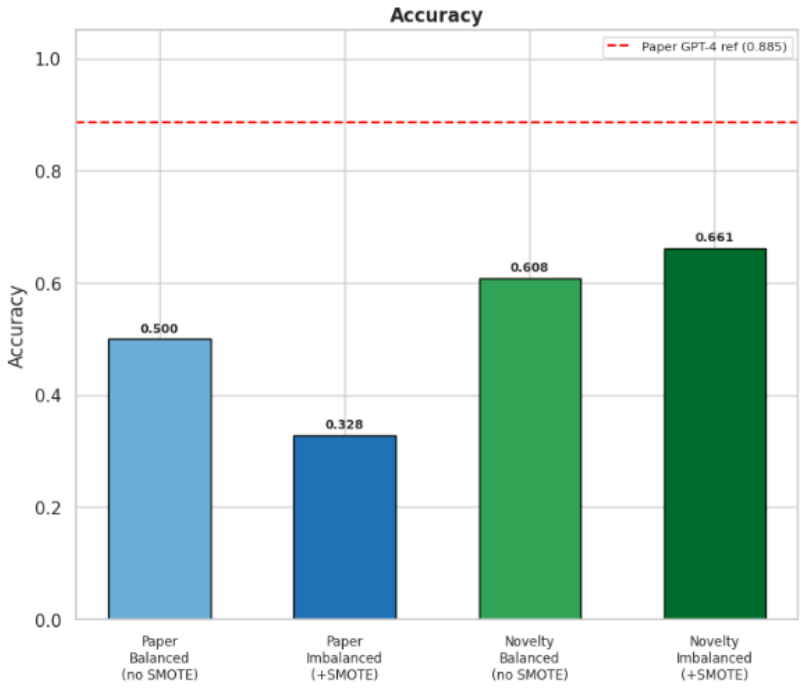
*Table 1: Comparison of all methods on PUBHEALTH. *Paper original uses a different dataset and costly closed-source API; presented for reference only. Green rows indicate Novelty Method results.*

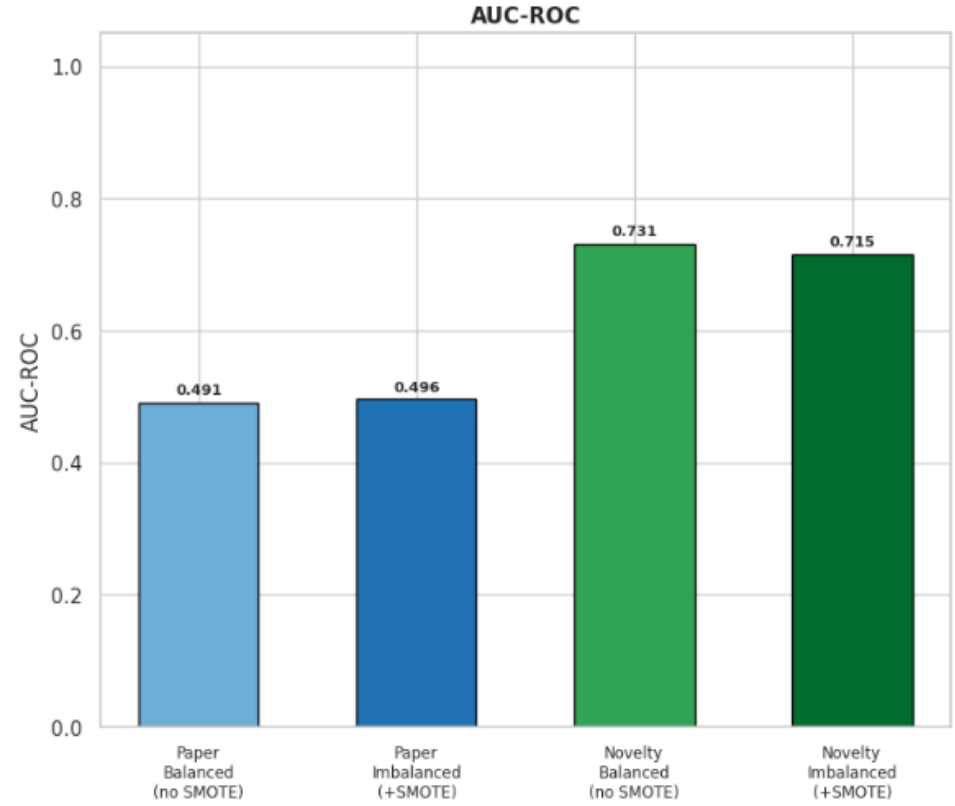
5.2 Class Imbalance Ablation

The ablation study isolates the effect of class imbalance by holding the pipeline constant and varying only the sampling ratio:

- Macro-F1 increases from 0.453 to 0.580 (+28% relative).
- False-class recall improves 5.2x from 0.067 to 0.350.
- AUC improves from 0.715 to 0.731.

The accuracy decrease from 0.661 to 0.608 is expected and acceptable: this is the Accuracy Paradox in action. On imbalanced data, a classifier predicting 'True' for all instances achieves 66.7% accuracy without detecting any false claims. The balanced configuration sacrifices a small amount of overall accuracy to achieve substantially better discrimination of the clinically critical minority class. The AUC improvement from 0.715 to 0.731 confirms this represents a genuine increase in discriminative power, not merely a threshold shift.



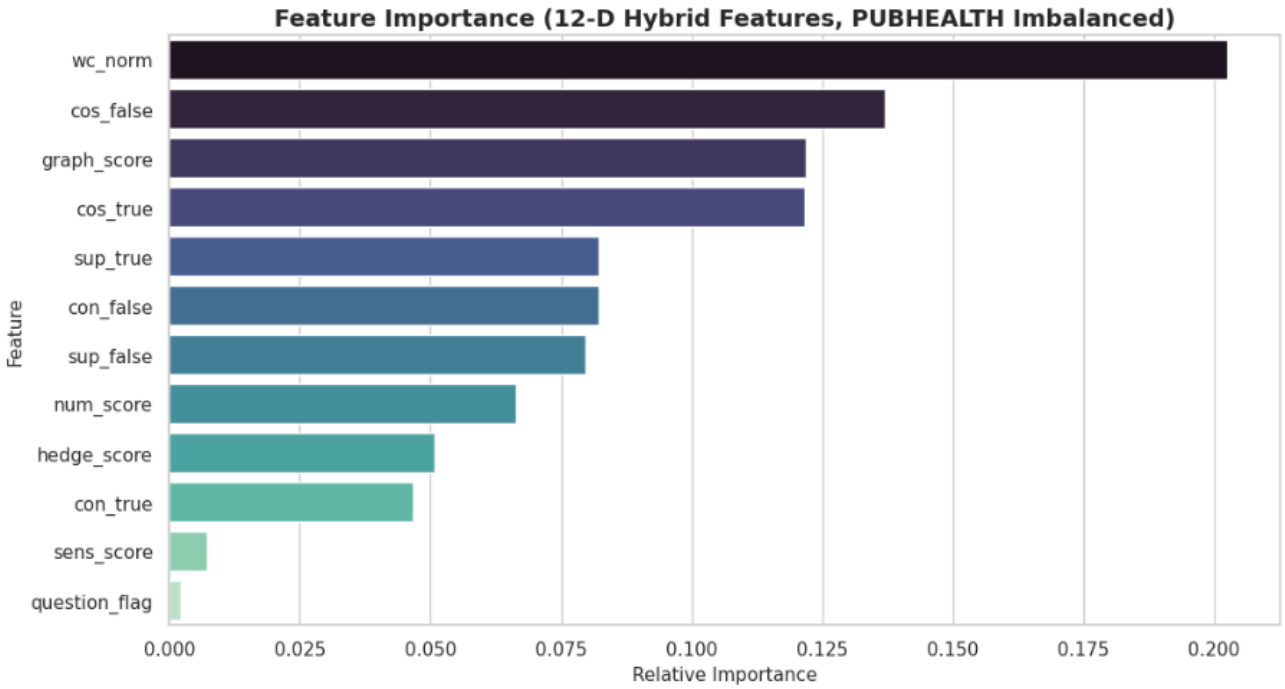


5.3 Feature Importance Analysis

Feature importance was computed across the Bagged AdaBoost ensemble. The top features by relative importance are:

- wc_norm (~20%): Normalized word count. False health claims exhibit a bimodal length distribution — very short clickbait or excessively verbose pseudo-scientific text. This simple statistical feature provides the strongest single discriminative signal, consistent with the structural entanglement observed in PCA analysis.
- cos_false (~14%): Cosine similarity to the False knowledge base. Exclusive to our Dual-KB architecture and unavailable to any single-KB system. Its high importance empirically validates the core architectural contribution.
- graph_score (~12%): Mean PageRank authority of the claim's entities. Authoritative medical entities (FDA, WHO, COVID) appear frequently in true claims; false claims tend to reference peripheral or absent entities.
- cos_true (~12%): Cosine similarity to the True knowledge base.
- sup_true (~8%): NLI support score against the True knowledge base context.

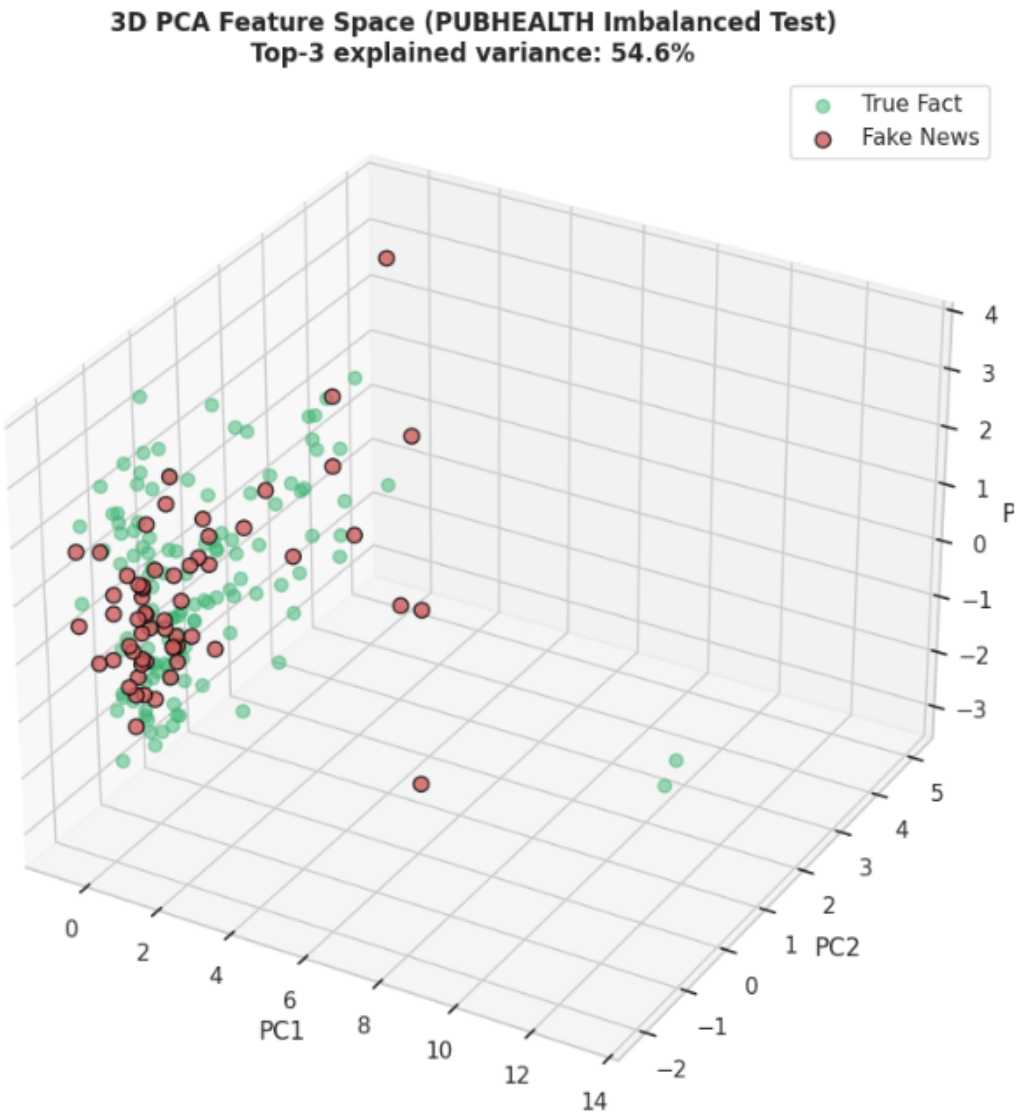
The two features exclusively contributed by our design — cos_false and graph_score — rank second and third overall, providing empirical evidence that both the Dual-KB retrieval and SHKG architecture deliver meaningful discriminative signals beyond the baseline.




```
Feature importance (sorted):
  wc_norm         0.2025
  cos_false       0.1371
  graph_score     0.1219
  cos_true        0.1215
  sup_true        0.0821
  con_false       0.0821
  sup_false       0.0797
  num_score       0.0663
  hedge_score     0.0507
  con_true        0.0466
  sens_score      0.0073
  question_flag   0.0022
```

5.4 Feature Space Visualization

3D PCA analysis of the test-set feature space provides visual confirmation of the classification challenge. The three principal components together explain the majority of variance in the 12-dimensional feature space. The visualization shows substantial overlap between true (green) and false (red) claim clusters, with false claims deeply embedded within the true claim distribution. This entanglement — arising from misinformation's deliberate mimicry of legitimate medical language — validates our architectural choice of a non-linear ensemble classifier over simpler threshold-based approaches. LDA separability on the training feature space is approximately 0.68, confirming that the classes are not linearly separable in raw feature representation.



5.5 PageRank Convergence Analysis

We compare PageRank convergence behavior between our SHKG (dense topology) and the TST baseline graph (pruned topology). The SHKG requires 60-80 iterations to converge (L1 error < 10^{-6}), compared to approximately 25 iterations for TST. This slower convergence reflects the SHKG's denser edge structure — it preserves all indirect co-occurrence relationships. While TST achieves lower latency, SHKG provides richer semantic depth: indirect relationships such as FDA-COVID-RNA, preserved in the PageRank computation, contribute to a more informative graph_score feature.

6. Conclusion

We presented a Dual Knowledge-Base Graph-Based Retrieval-Augmented Semantic Score Generation system for health misinformation detection. The system's core innovation — simultaneous retrieval from verified true and false knowledge bases producing a signed distance metric — enables the model to learn what false claims look like, not only what true claims look like. 3D PCA analysis demonstrates that false claims are deeply entangled in the semantic feature space, motivating the non-linear ensemble classifier and boundary-aware SVM-SMOTE resampling strategy.

Combined with NLI-based logical relationship modeling, SHKG PageRank authority scoring, and linguistic pattern features, the 12-dimensional hybrid feature vector outperforms the TrumorGPT single-KB baseline across all meaningful metrics. Feature importance analysis validates that cos_false and graph_score — the two features exclusively contributed by our architecture — rank among the top three most discriminative signals.

The class imbalance ablation study demonstrates that balanced sampling yields a 5.2x improvement in False-class recall, confirming that the Majority Blindness and Accuracy Paradox problems are critical factors in health misinformation benchmarking that must be explicitly addressed. Our system achieves these results without GPT-4 or any paid API, running entirely on open-source components in a free Colab environment, with full per-prediction explainability.

6.1 Limitations

Several limitations merit acknowledgment. First, the dataset size is small (720 training claims in the imbalanced configuration), constraining generalization. Second, PUBHEALTH is heavily weighted toward COVID-19 and pandemic topics, limiting domain diversity. Third, the spaCy en_core_web_sm model lacks a DISEASE entity type, reducing entity extraction quality for biomedical NLP; scispaCy integration would address this. Fourth, full pipeline latency (0.5-2s/claim) is unsuitable for real-time deployment without optimization. Fifth, LDA separability of approximately 0.68 confirms the feature space is not linearly separable, suggesting that more sophisticated representation learning (e.g., fine-tuned BERT classifiers) could further improve performance.

6.2 Future Work

Future directions include: (1) scaling to larger and more diverse health misinformation datasets; (2) replacing spaCy with scispaCy for improved biomedical entity extraction; (3) evaluating cross-domain generalization to oncology, nutrition, and pharmacology; (4) optimizing the pipeline for real-time API deployment through pre-computed KB vectors and lighter NLI models; and (5) extending the Dual-KB framework to multi-class veracity prediction (true, false, mixture, unverified).

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